Vadim Semenov

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EDUCATION

The University of Chicago
Ph.D. in Astronomy and Astrophysics 2019

Moscow Institute of Physics and Technology
B.S. & M.S. in Applied Mathematics and Physics *with honors* 2013

RESEARCH INTERESTS & EXPERIENCE

STEM Fellow | **Anthropic** since 07/2026

NASA Hubble and ITC postdoctoral fellow | **Harvard University** since 2019
ML/AI-aided subgrid models for supersonic turbulence simulations; large-scale cosmological simulations; physics of interstellar medium; early Universe; formation of galactic disks; numerical methods.

Research assistant | **The University of Chicago** | advisor: A. Kravtsov 2014 – 2019
Galaxy formation simulations; star formation and stellar feedback; cosmic rays; numerical methods

Research assistant | **Lebedev Physical Institute** | advisor: A. Doroshkevich 2010 – 2013
N-body simulations; large-scale structure of the Universe; properties of dark matter halos

PUBLICATIONS

36 papers in total; 14 first-author; 10 led by co-advised students. See the publication list below.
h-index: 20; total citations: >1200 (according to ADS, April 2026)
Links to ADS library, Google Scholar, and ORCID: 0000-0002-6648-7136

HONORS & AWARDS

Institute for Theory and Computation (ITC) Fellowship | Harvard 2022 – 2026
NASA Hubble Fellowship | Harvard 2019 – 2022
Schramm Fellowship | UChicago 2018 – 2019
Graduate Council Travel Fund Award | UChicago 2018
RCC Mind Bytes Award for Visualization in Research | RCC, UChicago 2015
McCormick Fellowship | UChicago 2013 – 2015
Vitaly Ginzburg Memorial Fellowship | Lebedev Physical Institute 2013
Dynasty Foundation Fellowship 2012
Abramov–Frolov Foundation Fellowship | MIPT 2008 – 2011

SELECTED TALKS

Invited colloquia and seminars
Astronomy Colloquium | UMass Amherst 04/2026
Astronomy Colloquium | UC Berkeley 02/2026
Origins Astronomy Seminar | MPA, Garching 07/2025
Astronomy Colloquium | University of Hawai'i 02/2025

Astronomy Colloquium | Johns Hopkins University 02/2025
Astronomy Colloquium | Harvard 01/2025
Astronomy Seminar | Tufts University, Boston 11/2024
CITA Seminar | CITA, Toronto 09/2023
Astronomy Colloquium | University of Michigan 11/2022
ISM Salon | CCA Flatiron Institute, NY 10/2022
Ringberg Seminar Series on Star Formation (virtual) 02/2021
Astronomy Colloquium | University of Maryland 02/2020
Cosmology Seminar | MPA, Garching 01/2019
CCAPP Seminar | OSU 11/2018
Galaxies & Cosmology Seminar | Harvard 10/2018
TAC Seminar | UC Berkeley 09/2018
Lunch Seminar | UW Madison 09/2018
Astro Seminar | Fermilab 09/2017

Conference Talks **⋆ – invited review** **† – invited talk**

†Revisiting the Hubble Sequence Across Cosmic Time | Cambridge, UK 04/2026
†Cosmic Turbulence and Magnetic Fields | Cargese, France 11/19; 09/23; 09/25
⋆12th Sino–German Meeting on Galaxy Formation | Chengdu, China 09/2025
⋆The role of feedback in galaxy formation | Potsdam, Germany 07/2025
Deciphering the cosmic code for galaxy formation | Puerto Varas, Chile 12/2024
†Celebrating 10 Years of the Illustris Project | Gargonza, Italy 10/2024
†Ringberg Workshop on Computational Galaxy Formation 05/16; 03/18; 04/22; 10/24
†Connecting Simulations and Observations in 3D | Barossa Valley, Australia 06/2024
⋆The Feedback-Driven Matter Cycle in Galaxies | Heidelberg, Germany 06/2024
GasPisa2024 | Pisa, Italy 05/2024
Building Galaxies from Scratch | Vienna, Austria 02/2024
Resolving Galaxy Ecosystems Across All Scales | Hong Kong, China 12/2023
⋆The Physics of Star Formation | Lyon, France 06/2023
Olympian Symposium – Star formation in the era of JWST | Katerini, Greece 06/2023
Star Formation in Different Environments | Quy Nhon, Vietnam 08/2022
†A Holistic View of Stellar Feedback and Galaxy Evolution | Ascona, Switzerland 07/2022
Midwest Magnetic Fields Meeting | UW Madison (virtual) 06/2021
ISM 2021: Structure, characteristic scales, and star formation | Beirut (virtual) 05/2021
Universality: Turbulence Across Vast Scales | Flatiron Institute, NY 12/2019
Harvard-Heidelberg Star Formation workshop 11/19; 12/22
10-th anniversary of KICC symposium | Cambridge, UK 09/2019
The Role of Feedback in Galaxy Formation | Potsdam, Germany 09/2018
Santa Cruz Galaxy Workshop | UC Santa Cruz 08/15; 08/16; 08/17; 08/18; 07/24; 08/25
The Role of Gas in Galaxy Dynamics | Valletta, Malta 10/2017
Carving through the Codes | Davos, Switzerland 02/2017

SUCCESSFUL PROPOSALS

PI, computing time on NASA HECC systems ~1.4M CPU hours 2021 – 2022
Modeling The Turbulent Evolution Of Galaxies Over Cosmic Time

Co-I, HST Cycle 33, 26 orbits (PI Ava Polzin) 2025
PANCHO: Piloting a Novel Calibration for H2 Observations

Co-I, ALMA 12m array ~40 hours (PI Eric Koch) 2023
Linking Molecular Cloud Structure to Massive Star Formation: 5000 molecular clouds, filaments, and bubbles across M33

Co-I, VLT MUSE ~20 hours (PI Anna McLeod) 2023
Resolving the environmental dependence of stellar feedback and massive star formation

CO-ADVISING STUDENTS

Sabrina Appel, former PhD student at Rutgers University (now NSF Fellow at AMNH)
Regulation of star formation in MHD simulations of ISM turbulence

Vedant Chandra, PhD student at Harvard
Formation of galactic disks in the TNG50 cosmological simulation

Jiwon Jesse Han, former PhD student at Harvard (now Stanford Science Fellow)
Connection between galactic disk warps and dark matter halos in TNG50

Ralf Konietzka, PhD student at Harvard
Modeling the dispersion measure of Fast Radio Bursts in cosmological simulations

Ava Polzin, PhD student at the University of Chicago
Modeling of molecular hydrogen in galaxy formation simulations

Mark Ugalino, PhD student at the University of Maryland
Comparing methods for modeling magnetic fields in galaxy simulations

Zuzanna Kocjan, PhD student at the University of Maryland
Tracing timescales of the star formation and feedback cycle in galaxy simulations

Rongrong Liu, PhD student at Harvard
Formation of galactic disks in galaxy simulations

Jasmine Palma, Undergraduate student at Harvard
Turbulent properties of intracluster media (senior thesis)

Zion Smith, Undergraduate at North Carolina Agricultural and Technical State University
Modeling the timescales of r-process enrichment (summer internship)

TEACHING EXPERIENCE

Teaching assistant | *Computational Astrophysics* (grad course) | UChicago 2016
Teaching assistant | *Stellar Astronomy and Astrophysics* (undergrad course) | UChicago 2013
Developed and taught *Introduction to Cosmology* (undergrad elective) | MIPT 2011 – 2012
Taught highschool-level courses and classes | Moscow 2007 – 2012

TECHNICAL SKILLS

Programming	C, C++, Python; MPI, OpenMP; git
Scientific computing	High-performance computing on NASA HECC and other clusters; large-scale cosmological galaxy simulations; numerical methods for astrophysics and hydrodynamics
ML/AI development	PyTorch training of custom NN architectures; C++/GPU integration for ML-aided numerical methods; custom skills and harnesses for agentic research workflows

LEADERSHIP & SERVICE

Committees	CfA Seminar organizing committee	2021 – 2024
	ITC Postdoctoral Fellowship Selection Committee	2023 – 2024
	CGM workshop "Multiphase Madness", CfA	2024
	ITC Colloquium co-organizer	since 2024
Peer reviews	Journals: Nature, ApJ, MNRAS, A&A, LRCA	
	Expert reviewer for NASA FINESST Fellowship	2020 – 2021
	Expert reviewer for NASA HST cycles 32 and 33	2024 – 2025
	Expert reviewer for BSF grants	2023 – 2025

APPEARANCES IN MEDIA

Scientific American April 2025
"JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History"

Forbes June 2023
"Our Milky Way Is Somewhat Of An Oddball Galaxy, New Study Confirms"

REFERENCES

Andrey Kravtsov (PhD advisor)	UChicago | kravtsov@uchicago.edu
Charlie Conroy	Harvard | cconroy@cfa.harvard.edu
Lars Hernquist	Harvard | lhernquist@cfa.harvard.edu
Nick Gnedin	Fermilab | gnedin@fnal.gov
Romain Teyssier	Princeton | teyssier@princeton.edu

PUBLICATION LIST

h-index: 20; total citations: >1200 (according to ADS, April 2026)

First author (14 total; ADS link)

1. **Semenov** \| ApJ 1010, 123 \| arXiv:2602.04950 2026
   How Do Disk Galaxies Form?
2. **Semenov**, Conroy, Hernquist \| ApJ 989, 219 \| arXiv:2410.09205 2025
   From UV-bright Galaxies to Early Disks: the Importance of Turbulent Star Formation in the Early Universe
3. **Semenov**, Conroy, Smith, Puchwein, Hernquist \| ApJ 990, 7 \| arXiv:2409.18173 2025
   How Early Could the Milky Way's Disk Form?
4. **Semenov** \| ApJS 281, 37 \| arXiv:2410.23339 2025
   Capturing Turbulence with Numerical Dissipation: a Simple Dynamical Model for Unresolved Turbulence in Hydrodynamic Simulations
5. **Semenov**, Conroy, Chandra, Hernquist, Nelson \| ApJ 962, 84 \| arXiv:2306.09398 2024
   Formation of Galactic Disks I: Why did the Milky Way's Disk Form Unusually Early?
6. **Semenov**, Conroy, Chandra, Hernquist, Nelson \| ApJ 972, 73 \| arXiv:2306.13125 2024
   Formation of Galactic Disks II: the Physical Drivers of Disk Spin-up
7. **Semenov**, Kravtsov, Diemer \| ApJS 261, 16 \| arXiv:2107.14240 2022
   Entropy-Conserving Scheme for Modeling Nonthermal Energies in Fluid Dynamics Simulations

8. **Semenov**, Kravtsov, Gnedin | ApJ 918, 13 | arXiv:2103.13406 2021
Spatial Decorrelation of Young Stars and Dense Gas as a Probe of the Star Formation–Feedback Cycle in Galaxies
9. **Semenov**, Kravtsov, Caprioli | ApJ 910, 126 | arXiv:2012.01427 2021
Cosmic Ray Diffusion Suppression in Star-Forming Region Inhibits Clump Formation in Gas-Rich Galaxies
10. **Semenov**, Kravtsov, Gnedin | ApJ 870, 79 | arXiv:1809.07328 2019
What Sets the Slope of the Molecular Kennicutt–Schmidt Relation?
11. **Semenov**, Kravtsov, Gnedin | ApJ 861, 4 | arXiv:1803.00007 2018
How Galaxies Form Stars: the Connection between Local and Global Star Formation in Galaxy Simulations
12. **Semenov**, Kravtsov, Gnedin | ApJ 845, 133 | arXiv:1704.04239 2017
The Physical Origin of Long Gas Depletion Times in Galaxies
13. **Semenov**, Kravtsov, Gnedin | ApJ 826, 200 | arXiv:1512.03101 2016
Nonuniversal Star Formation Efficiency in Turbulent ISM
14. **Semenov** | Astronomy Reports 57, 485 2013
Statistical Analysis of the Large-Scale Structure of the Universe Using Observational Data and Numerical Modeling

Led by co-advised students (10 total; ADS link)

15. Ugalino, Diemer, **Semenov**, Kim, et al. | ApJ subm. | arXiv:2610.02308 2026
Towards robust simulations of the galactic dynamo: a cross-code comparison between Eulerian and Lagrangian schemes
16. Kocjan, Diemer, **Semenov**, Bialy, Malamud | ApJ subm. | arXiv:2608.28747 2026
Shock-heated Away: The Impact of Radiative Cooling on Gas-Phase Transitions in Supernova Remnants
17. Kocjan, **Semenov** | ApJ 1007, 46 | arXiv:2602.02657 2026
The Rhythm of the ISM: Tracing the Timescales of Gas Evolution and Star Formation across Galactic Environments
18. Konietzka, Connor, **Semenov**, et al. | ApJ accepted | arXiv:2507.07090 2025
Ray-tracing Fast Radio Bursts Through IllustrisTNG: Cosmological Dispersion Measures from Redshift 0 to 5.5
19. Polzin, Kravtsov, **Semenov**, Gnedin | OJAp 7, 114 | arXiv:2407.11125 2024
On the universality of star formation efficiency in galaxies
20. Chandra, **Semenov**, Rix, Conroy, et al. | ApJ 972, 112 | arXiv:2310.13050 2024
The Three-Phase Evolution of the Milky Way
21. Polzin, Kravtsov, **Semenov**, Gnedin | ApJ 966, 172 | arXiv:2310.10712 2024
Modeling Molecular Hydrogen in Low Metallicity Galaxies
22. Han, **Semenov**, Conroy, Hernquist | ApJL 957, L24 | arXiv:2309.07208 2023
Tilted Dark Halos are Common, Long-Lived, and Can Warp Galactic Disks
23. Appel, Burkhart, **Semenov**, Federrath, Rosen, Tan | ApJ 954, 93 | arXiv:2301.07723 2023
What Sets the Star Formation Rate of Molecular Clouds? The Density Distribution as a Fingerprint of Compression and Expansion Rates

24. Appel, Burkhart, **Semenov**, Federrath, Rosen | ApJ 927, 75 | arXiv:2109.13271 2022
The Effects of Magnetic Fields and Outflow Feedback on the Shape and Evolution of the Density PDF in Turbulent Star-Forming Clouds

In collaboration (12 total)

25. Wheeler, Kravtsov, Chiti, Katz, **Semenov** | OJAp 8, 151 | arXiv:2507.03182 2025
What Sets the Metallicity of Ultra-Faint Dwarfs?

26. Segovia Otero et al. | MNRAS 538, 2646 | arXiv:2410.08266 2025
Cosmic evolution of the star formation efficiency in Milky Way-like galaxies

27. Robinson, Avestruz, Gnedin, **Semenov** | OJAp subm. | arXiv:2412.15324 2024
The effects of different cooling and heating function models on a simulated analog of NGC300

28. Beane, Johnson, **Semenov**, et al. | ApJ 985, 221 | arXiv:2410.21580 2024
Rising from the Ashes II: The Bar-driven Abundance Bimodality of the Milky Way

29. Aung et al. | MNRAS 532, 2965 | arXiv:2403.00912 2024
Entrainment of Hot Gas into Cold Streams: The Origin of Excessive Star-formation Rates at Cosmic Noon

30. Jeffreson, **Semenov**, Krumholz | MNRAS 527, 7093 | arXiv:2301.10251 2024
Clouds of Theseus: long-lived molecular clouds are composed of short-lived H_2 molecules

31. Kiihne, Appel, Burkhart, **Semenov**, Federrath | ApJ 979, 89 | arXiv:2305.11218 2023
Fitting Probability Distribution Functions in Turbulent Star-Forming Molecular Clouds

32. Shin, Tacchella, Kim, Iyer, **Semenov** | ApJ 947, 61 | arXiv:2211.01922 2023
Star Formation Variability as a Probe for the Baryon Cycle within Galaxies

33. Bialy, Zucker, Goodman, Foley, Alves, **Semenov**, Benjamin, Leike, Enßlin
ApJL 919 L5 | arXiv:2109.09763 2021
The Per-Tau Shell: A Giant Star-forming Spherical Shell Revealed by 3D Dust Observations

34. Gnedin, **Semenov**, Kravtsov | JCoPh 359, 93 | arXiv:1801.03108 2018
Enforcing the Courant–Friedrichs–Lewy Condition in Explicitly Conservative Local Time Stepping Schemes

35. Li, Gnedin, Gnedin, Meng, **Semenov**, Kravtsov | ApJ 834, 69 | arXiv:1608.03244 2017
Star Cluster Formation in Cosmological Simulations. I. Properties of Young Clusters

36. Kim et al. (AGORA Collaboration) | ApJ 833, 202 | arXiv:1610.03066 2016
The AGORA High-resolution Galaxy Simulations Comparison Project. II. Isolated Disk Test